Question	Answer	Marks	Partial Marks
8(a)(i)	1200	**1**	
8(a)(ii)(a)	800	**3**	**M2** for [2 ×] (20 × 12 + 20 × 5 + 12 × 5) or **M1** for 20 × 12 or 20 × 5 or 12 × 5
8(a)(ii)(b)	0.19	**1**	**FT** 152 ÷ *their* 800
8(b)	$\frac{3x}{2}$ or 1.5x	**3**	**B2** for $r^3 = \frac{27x^3[\pi]}{8[\pi]}$ or better or **M1** for $\frac{4}{3}\pi r^3 = \pi x^2 \times \frac{9x}{2}$
8(c)	13.6 or 13.59 to 13.61	**7**	If chord is *AB* and *O* is centre of the cross section **M2** for 2 × $\cos^{-1}\left(\frac{20-5}{20}\right)$ oe or **M1** for cos = $\frac{20-5}{20}$ oe **M1** for $\frac{theirAOB}{360} \times \pi \times 20^2$ or $\frac{1}{2}(20)^2\left(\frac{82.8\pi}{180}\right)$ **M1** for $\frac{1}{2} \times 20^2 \times \sin(their\ AOB)$ oe **M1** for *their* area × 150 **M1** for *their* volume ÷ 1000
9(a)	42.3 or 42.28 to 42.30...	**7**	**M1** for $\frac{AB}{14} = \cos 35$ oe **M1** for $\frac{AD}{14} = \sin 35$ oe **B1** for [C =] 75 **M3** for [BC =] $\frac{14\sin 60}{\sin their 75}$ oe **and** [DC] $\frac{14\sin 45}{\sin their 75}$ oe or **M2** for $\frac{14\sin 60}{\sin their 75}$ **or** $\frac{14\sin 45}{\sin their 75}$ oe or **M1** for $\frac{\sin their 75}{14} = \frac{\sin 60}{BC}$ oe or $\frac{\sin their 75}{14} = \frac{\sin 45}{CD}$ oe

Question	Answer	Marks	Partial Marks
9(b)(i)	4.91 or 4.907...	**3**	**B2** for [l^2 =] 24.1 or 24.08... or M2 for $\sqrt{3}\,l = 8.5$ or [l =] $\sqrt{\frac{8.5^2}{3}}$ oe or **M1** for $l^2 + l^2 + l^2 = 8.5^2$ oe
9(b)(ii)	35.3 or 35.26 to 35.3 nfww	**3**	**M2dep** for sin (angle) = $\frac{\textit{their}\ \textbf{(b)(i)}}{8.5}$ oe or **M1** for clear recognition of correct angle
10(a)(i)	4	**1**	
10(a)(ii)	3	**1**	
10(a)(iii)	13	**1**	**FT** 5 × *their* **(a)(i)** – 7
10(b)	$\frac{x+2}{3}$ final answer	**2**	**M1** for $y + 2 = 3x$ or for $\frac{y}{3} = x - \frac{2}{3}$ or for $x = 3y - 2$
10(c)	$9x^2 - 9x + 2$ final answer	**3**	**M1** for $(3x-2)^2 + 3x - 2$ **B1** for $(3x-2)^2 = 9x^2 - 6x - 6x + 4$
10(d)	$2x + 1$	**1**	
10(e)(i)	81	**1**	
10(e)(ii)	x	**1**	Not $y = x$
11(a)(i)	–5	**1**	
11(a)(ii)	Subtract 4 oe	**1**	
11(a)(iii)	$15 - 4n$ oe final answer	**2**	**B1** for $k - 4n$ or $15 - jn$ $j \neq 0$
11(b)(i)	$\frac{1}{21}$ or equivalent fraction	**2**	**B1** for $\frac{12}{7}$ and $\frac{10}{6}$
11(b)(ii)	$n = \frac{3}{5}$ oe or $2n \geqslant n + 1$ but $3 < 4$.	**M2**	**M1** for $\frac{3}{4} = \frac{2n}{n+1}$ oe or **M1** for $2n > n + 1$ but $3 < 4$
	No, n is not an integer oe or No, $\frac{3}{4}$ is less than 1, oe	**A1**	